

## CEMA-3 Flue Gas Analyzer

### Product Description

CEMA-3 flue gas analyzer is a self-developed gas analysis product for domestic and overseas environmental and industrial on-line monitoring online analysis. The analyzer is based on DOAS and chemometric algorithms (PLS), able to measure SO<sub>2</sub>, NO, NO<sub>2</sub>, O<sub>2</sub>, NH<sub>3</sub>, CO, CO<sub>2</sub>, Cl<sub>2</sub>, O<sub>3</sub>, H<sub>2</sub>S, HCl, CH<sub>3</sub>I, etc.. With high accuracy and reliability, low operating costs, fast response time, wide measurement range and application fields, it has achieved even surpassed similar products home and abroad. It can be widely used in environmental online monitoring, industrial process control, safety monitoring, etc.

### Application

Flue gas measuring devices are applied in monitoring flue gas emissions from coal-fired thermal power plants, garbage burning power plants, glass factories, lime plants, ceramics, furnaces, chemical industry, ...

### Features

- Multi-return gas chamber
- Low detection limit and temperature-drift
- Getting NO<sub>x</sub> by directly measuring NO and NO without NO→NO<sub>2</sub> converter
- No interference of particulate matter, moisture and other factors
- No optical moving parts, high reliability, no interference to the measurement by field.

### Technical Data

|                       |                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| Measuring Technique   | DOAS UV (SO <sub>2</sub> , NO <sub>x</sub> ), Electrochemistry (CO, O <sub>2</sub> )                   |
| Measuring Range       | 0-1600 ppm (NO <sub>x</sub> ), 0-1200 ppm (SO <sub>2</sub> ), 0-25% (O <sub>2</sub> ), 0-2700 ppm (CO) |
| Linearity Error       | ≤ ±1.5% F.S.                                                                                           |
| Repeatability         | ≤ ±1.0% F.S.                                                                                           |
| Zero Drifting         | ≤ ±1.5% F.S/24h                                                                                        |
| Measuring Range Drift | ≤ ±2% F.S/7d                                                                                           |
| Response Time         | < 90s (T <sub>90</sub> )                                                                               |
| Warm-up Time          | None                                                                                                   |
| Relay Output          | 14                                                                                                     |
| Binary Input          | 6                                                                                                      |
| Analog Output         | 4x4 – 20mA                                                                                             |
| Analog Input          | 2x4 – 20mA                                                                                             |
| Digital Communication | 1xRS232/RS485                                                                                          |
| Enclosure Rating      | IP65                                                                                                   |
| Ambient Temperature   | -10°C – 40°C                                                                                           |
| Ambient Humidity      | 0%Rh – 95%Rh                                                                                           |

### CEMA-3

